Lab 7 Reading Guides

These are optional guides that might help your readings of the paper. You do not have to turn these in. Jump to the right reading guide for your paper.

1) **CRICKET PILLOW TALK (Hagg et al., 2024)**

Glossary
Oviposition: laying eggs
Spermatophore: a large capsule full of sperm and nutrients. In some species (including crickets), the male deposits the spermatophore externally onto the female.
Phonotactic behavior: movement in response to a sound. Animals can be positively phonotactic (approach the source of the sound) or negatively phonotactic (go away from the sound).
Eclosion: emergence of the final adult stage of the cricket
Pronotum: the dorsal (top) plate of an insect's first thorax segment
Sperm competition: competition between the sperm of two or more males to fertilize one female's egg(s)
Plastic behavior: a behavior that changes in response to an individual's environment

Key Questions (answer them and you'll get the gist of the paper):

a) When a male animal performs mate guarding behavior after copulation, do both the male and female benefit? Why or why not?
b) In some species of insects, males guard mates by physically restraining them. Why is this not observed in crickets?
c) Why would male crickets benefit from mate guarding? (I.e., what do some female crickets do after mating?)
d) Why would it be unusual for a male cricket to sing immediately after mating?
e) In the emerald coral goby (a tropical fish), one male lives in a coral burrow with multiple females. However, a study by Wong et al. (2008) repeatedly shows that males are monogamous and will only mate with the same female. What do you think is happening in this animal system?

Tips
None! This paper is a straightforward read.

2) BUMBLEBEE PLAYING (Galpayage Dona et al., 2022)

Glossary

Conspecific: an individual of the same species
Affective state: longer lasting mood states of an animal. Positive examples: excited, relaxed. Negative examples: Bored, fearful.
Ad libitum: (of food) Unrestricted, unlimited.
Honeypot: one of many sphere-shaped structures in a bumble bee hive where nectar is stored
GLM: Generalized linear model. A statistical model that coerces non-parametric data into a normal distribution.

Key Questions: _(answer them and you'll get the gist of the paper):_

a) If we changed Experiments 1 & 2 to include balls in the unobstructed path (Fig 1), how would that invalidate the findings of the experiment?
b) Why did the researchers perform Experiment 3 with different colored chambers?
c) What's the point of measuring how the bees moved the balls (like in Fig 6c)?
d) Let's say the researchers used yellow cotton balls instead of wooden ones. Why would this be a big flaw in the design?
e) According to the established definition of "play behavior," which of the following dogs shows an example of genuine play behavior?
    - *Luna*: she rolls over, but only when you give her a treat.
    - *Rocky*: as a young dog, he used to beg you to play fetch all the time. Now that he's 17, he would rather sleep all day.
    - *Archie*: he loves digging holes in the yard for no discernable reason.

Tips

If the methods of this study don't immediately make sense to you, take a look at the figures and read the results. They will help you visualize why the researchers chose to perform each experiment in the way they did.

3) OCTOPUS AGING (Roura et al., 2024)

Glossary
Supra-esophageal brain (ganglia): Mass of octopus brain above the esophagus, the other part is the sub-esophageal mass.
Brooding: protection and oxygenation of eggs
Spermatophore/spermatheca: a large capsule full of sperm and nutrients.
Chromatophore: special pigment cells that allow octopuses to change color and blend with their surroundings

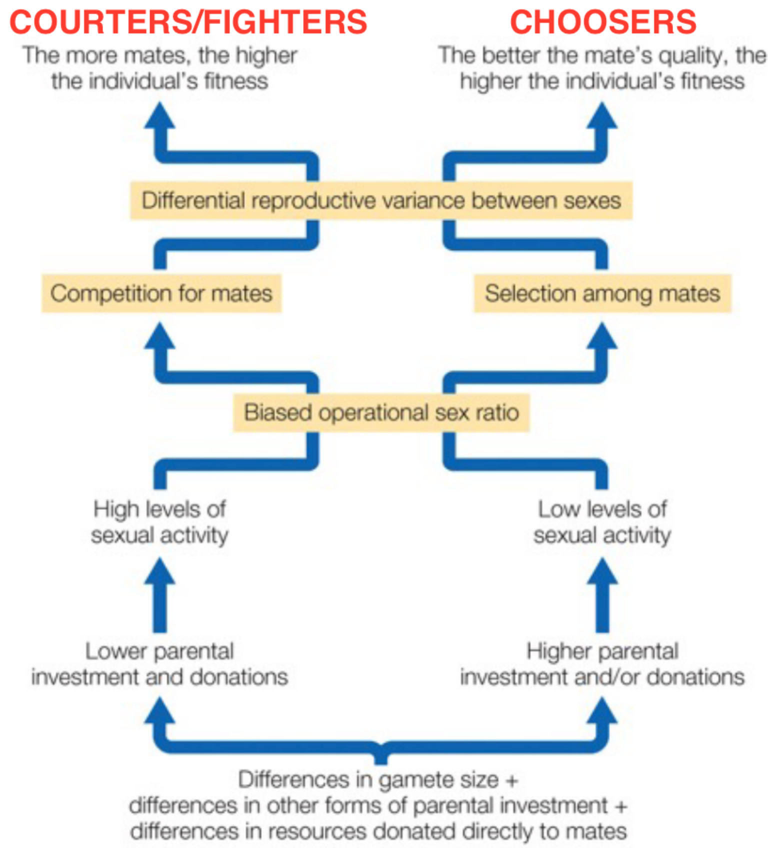
Key questions (answer them and you'll get the gist of the paper):

a) Decide which animals are semelparous and which ones are iteroparous.
    - *Salmon*. To mate, sexually mature adults return to the stream where they hatched. Dams often have "fish ladders" so that the salmon can climb up them, and biologists note that there is much more traffic going up than down.
    - *Sea turtles.* Female sea turtles also return to their birthplace to lay eggs, often swimming hundreds or thousands of miles. Famously, the New England Aquarium has kept Myrtle the green sea turtle since 1970.
    - *Gorillas.* Along with humans and chimpanzees, gorillas are one of the few mammal species that naturally get grey hair as they age.
    - *Mayflies.* Like many insects, mayflies undergo incomplete metamorphosis. They spend most of their lives underwater and emerge as winged adults. On some summer days, it's common to see 1000's of mayflies flying in the air.

b) The paper supports a long-known fact that female octopi starve themselves to death while protecting their eggs. What happens when you remove the optic gland of a female octopus that laid eggs? What does that say about this behavior?

c) As they approach death, all of the males' arms *except* the hectocotylus (third right arm, Figure 5) degenerate. How does this support the behavior the researchers observed in males?

d) The study finds that before death, females become less active to protect their eggs, while males exhibit more mate finding behavior. To the right are the two general sexual strategies we discussed in lecture. In octopi, which sex is the “chooser,” and which sex is the “courter”?



ANIMAL BEHAVIOR 11e, Figure 9.6

<u>Tips</u>

There are quite a few anatomical terms in this paper, some of which are specific to octopi. Ctrl/Cmd+F the term if you are lost– the authors generally do a good job of clarifying what each term means. Terms that are not defined well are in the Glossary.

4) WHALE CULTURE (Filatova et al., 2024)

Glossary

Behavioral repertoire: all of an animal's behaviors
Rorqual: a family of baleen whales (no teeth) that includes the blue whale
Fission-fusion society: a group of social animals that "forms, breaks up, and reforms at frequent intervals," per Conrad & Roper (2000)
Shelf break: The edge of the continental shelf and thus the end of shallow water
Haplotype: a group of alleles that are collectively inherited from the same parent
Horizontal transmission: (of behavior) learned, not inherited
Trophic niche: the specific set of conditions where an animal has enough food to survive

Key Questions (answer them and you'll get the gist of the paper):

a) Of all the interesting behaviors that the researchers could've picked, why did they choose shallow water usage as their study question? Based on this answer, do you think cultural behaviors in beaked whales are truly rare?
b) Why did the researchers conduct a genetic analysis of the whales? (And how did they collect samples?) What would it mean if all the whales using shallow water were genetically closely related?
c) Look at Figure 2. The resident and mixed groups (2a &2b) clearly cluster in the shallow water near the island, while transients (2c) hang out in deeper waters. But there are a small group of transients seen near the island too. What would figure 2c look like if this study ran for a few more years?
d) Now that you know what cultural behavior in animals is, which of the following is an example of culture?
    - Your neighbors open a catering business. Raccoons have started hanging out between your houses because of all the food waste in their trash cans.
    - A family group of geese at Campus Pond have claimed a corner of the pond for themselves. Their wings are unusually flexible, so they can successfully scare away other geese with a unique wing-flapping motion.
    - Your aunt is a cat lover who always owns three cats at once. She tells you that for all her life, every one of her cats has had the strange ability to open doorknobs.

Tips

Don't get discouraged by the second statistical analysis of this paper (NBDA/OADA):

1) The NBDA/OADA analysis is trying to figure out if using shallow water is a behavior that is learned from copying another individual.
2) The researchers built a social network (Figure 1) to assess how each whale associates with the other whales.
3) If closely associated individuals in the social network all know the "shallow water behavior" while those unassociated with them do not, it is likely that the behavior is learned socially. **This is what the analysis found**.

    There's some other stuff going on behind the model – If you are interested in following the math behind it, the creators of this analysis describe it in Hoppitt et al. (2010)